

Csv's uploaded

The screenshot shows the Databricks Catalog Explorer interface. The left sidebar contains a navigation menu with options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover Beta, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, and SQL Warehouses. The main panel displays the path `/Volumes/vita_catalog/dataengineering/data`. A table lists the files in this directory:

Name	Size	Last modified
NYhost.csv	5.69 MB	3 days ago
Online Retail.csv	42.96 MB	2 days ago
SalesTraining.csv	114.84 KB	3 days ago
airports.csv	10.41 MB	44 seconds ago
case_url.csv	71.20 KB	4 days ago
module_8_students_data.csv	317.00 B	5 days ago
module_8_students_data.txt	317.00 B	5 days ago
runways.csv	3.20 MB	46 seconds ago
train.csv	25.67 MB	4 days ago

The screenshot shows the Databricks Notebook editor interface. The left sidebar contains a navigation menu with options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover Beta, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, and SQL Warehouses. The main panel displays a Python script that reads CSV files into Spark DataFrames:

```
1 #Read CSV Files into Spark DataFrames
2 df_airports = spark.read.csv(
3     "/Volumes/vita_catalog/dataengineering/data/airports.csv",
4     header=True,
5     inferSchema=True
6 )
7
8 df_runways = spark.read.csv(
9     "/Volumes/vita_catalog/dataengineering/data/runways.csv",
10    header=True,
11    inferSchema=True
12 )
13
14 df_airports.printSchema()
15 df_runways.printSchema()
16
17 df_airports.show(5)
18 df_runways.show(5)
```

The output of the script shows the schema for both DataFrames:

```
root
|-- id: integer (nullable = true)
|-- ident: string (nullable = true)
|-- type: string (nullable = true)
|-- name: string (nullable = true)
```

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Workspace

Recents

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Queries

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Data Engineering

Runs

Data Ingestion

New Notebook 2026-05-09 12:33:36 × +

File Edit View Run Help Python ☆ Last edit was now

Python 2

Just now (1s)

df_combined.show(5)

See performance (1)

df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 37 more fields]

id ident	type	name latitude_deg longitude_deg elevation_ft continent iso_country iso_region municipality scheduled_se	
rvice icao_code iata_code gps_code local_code	home_link wikipedia_link keywords	id airport_ref airport_id length_ft width_ft surf	
ac lighted closed le_ident le_latitude_deg le_longitude_deg le_elevation_ft le_heading_degT le_displaced_threshold_ft	he_ident he_latitude_deg he_	longitude_deg he_elevation_ft he_heading_degT he_displaced_threshold_ft	
6523 00A heliport Total RF Heliport 40.070985 -74.933689 11 NA US US-PA Bensalem	no NULL NULL K00A 00A https://www.pennd... NULL NULL 269408 6523 00A 80 ASPH-G	1 0 H1 NULL NULL NULL NULL NULL NULL NULL	
6524 00AK small airport Lowell Field 59.947733 -151.692524 450 NA US US-AK Anchor Point	no NULL NULL 00AK 00AK NULL NULL 255155 6524 00AK 2500 40 GRVL	0 0 N NULL NULL NULL NULL S NULL NULL	
6525 00AL small airport Epos Airpark 34.8647995 -86.77030182 820 NA US US-AL Harvest			

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workspace

New Notebook 2026-05-09 12:33:36

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Run all Serverless Schedule Share

```
1 #Add runway category column
2 from pyspark.sql.functions import when, col
3
4 df_combined = df_combined.withColumn(
5     "runway_category",
6     when(col("length_ft") >= 10000, "Long")
7     .when((col("length_ft") >= 5000) & (col("length_ft") < 10000), "Medium")
8     .otherwise("Short")
9 )
10
11 df_combined.select(
12     "name",
13     "length_ft",
14     "runway_category"
15 ).show(10)
```

> See performance (1)

df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 38 more fields]

	name	length_ft	runway_category
1	Total RF Heliport	80	Short
2	Lowell Field	2500	Short
3	Epps Airpark	2100	Short
4	Katmai Lodge Airport	4517	Short
5	Fulton Airport	1450	Short
6	Cordes Airport	1700	Short
7	Goldstone (GTS) A...	6000	Medium
8	Grass Patch Airport	3200	Short

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```
1 #Temporary View
2 df_combined.createOrReplaceTempView("airports_view")
```

> See performance (1)

Part C

Top10 countries

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workspace

New Notebook 2026-05-09 12:33:36

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Run all Serverless Schedule Share

```
1 %sql
2 SELECT
3     iso_country,
4     COUNT(*) AS runway_count
5 FROM airports_view
6 GROUP BY iso_country
7 ORDER BY runway_count DESC
8 LIMIT 10
```

> See performance (1) Optimize

_sqlidf: pyspark.sql.connect.dataframe.DataFrame = [iso_country: string, runway_count: long]

	iso_country	runway_count
1	US	5825
2	BR	2103
3	AU	1578
4	CA	761
5	AR	659
6	FR	600
7	GB	592
8	DE	489
9	RU	414
10	ID	

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Average Runway Length

The screenshot shows a Databricks workspace with a new notebook titled "New Notebook 2026-05-09 12:33:36". The notebook contains a SQL query that calculates the average runway length for each continent, excluding null values. The query is as follows:

```
1 %sql
2 SELECT
3     continent,
4     AVG(length_ft) AS avg_runway_length
5 FROM airports_view
6 WHERE length_ft IS NOT NULL
7 GROUP BY continent
8 ORDER BY avg_runway_length DESC
```

The query has been executed, and the results are displayed in a table with 7 rows. The table has two columns: "continent" and "avg_runway_length". The results are as follows:

continent	avg_runway_length
AN	7768.857142857143
AS	6925.6469996647675
AF	6787.526448362721
EU	4103.151335311572
OC	3558.0401785714284
SA	2857.0891141239963
NA	2599.724218941758

The table is refreshed now. The status bar at the bottom shows 7 rows and 1.71s runtime.

Lighted Runway

The screenshot shows a Databricks workspace with a new notebook titled "New Notebook 2026-05-09 12:33:36". The notebook contains a SQL query that counts the number of lighted runways for each airport, filtered by country and lighted status. The query is as follows:

```
1 %sql
2 SELECT
3     name,
4     municipality,
5     COUNT(*) AS lighted_runways
6 FROM airports_view
7 WHERE iso_country = 'IN'
8 AND lighted = 1
9 GROUP BY name, municipality
10 ORDER BY lighted_runways DESC
```

The query has been executed, and the results are displayed in a table with 9 rows. The table has three columns: "name", "municipality", and "lighted_runways". The results are as follows:

name	municipality	lighted_runways
Indira Gandhi International Airport	New Delhi	3
Navi Mumbai International Airport	Navi Mumbai	2
Kempegowda International Airport Bengaluru	Bengaluru	2
Mangaluru International Airport	Mangaluru	2
Chennai International Airport	Chennai	2
Ambala Air Force Station	Ambala	2
Bidar Airport / Bidar Air Force Station	Bidar	2
Agra Airport / Agra Air Force Station	Agra	2
Rajiv Gandhi International Airport	Hyderabad	2

The table is refreshed now. The status bar at the bottom shows 9 rows and 1.71s runtime.

Top 5 Airport with longest max runway length

The screenshot shows a Databricks workspace with a new notebook. The SQL query is as follows:

```
1 %sql
2 SELECT
3   name,
4   iso_country,
5   MAX(length_ft) AS max_runway_length
6 FROM airports_view
7 GROUP BY name, iso_country
8 ORDER BY max_runway_length DESC
9 LIMIT 5
```

The results of the query are displayed in a table with 5 rows and 3 columns: name, iso_country, and max_runway_length.

	name	iso_country	max_runway_length
1	Gunflint Seaplane Base	US	30000
2	Libby Camps Seaplane Base	US	26000
3	Brookville Reservoir Seaplane Ba...	US	25000
4	Long Lake Seaplane Base	US	25000
5	Conchas Lake Seaplane Base	US	21120

The table is refreshed now. The runtime for the query is 1.81s.

Save as Parquet

The screenshot shows a Databricks workspace with a new notebook. The Python code is as follows:

```
1 df_combined = df_combined.drop(df_runways.id)
2 df_combined.write.mode("overwrite").parquet(
3   "/Volumes/vita_catalog/dataengineering/data/airports_runways_parquet"
4 )
```

The code is executed, and the results are displayed in a table with 1 row and 1 column: Start typing or generate with AI (Ctrl + I)...

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1	Start typing or generate with AI (Ctrl + I)...

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